

Apriori Association Rules: Numerical Walkthrough

The Apriori algorithm is used for Market Basket Analysis to discover purchase relationships between products. It operates on the principle that: *If an itemset is frequent, then all of its subsets must also be frequent.* This allows the algorithm to prune (discard) unpopular item combinations early, saving massive amounts of computation. It evaluates rules using **Support**, **Confidence**, and **Lift**.

The Sample Dataset

Imagine our supermarket database has exactly 3 customer receipts (transactions):

- **Receipt 1 (T1):** {Bread, Butter}
- **Receipt 2 (T2):** {Bread, Butter, Milk}
- **Receipt 3 (T3):** {Milk}

We configure Apriori with a **Minimum Support of 50%** (must appear in ≥ 2 receipts) and a **Minimum Confidence of 60%**.

Step-by-Step Association Mining

Step 1: Candidate Single Items (\$C_1\$)

We list individual items and count their support:

- **{Bread}**: 2 receipts (T1, T2) → Support = 66.7% (Passes!)
- **{Butter}**: 2 receipts (T1, T2) → Support = 66.7% (Passes!)
- **{Milk}**: 2 receipts (T2, T3) → Support = 66.7% (Passes!)

All items become **Frequent 1-Itemsets (\$L_1\$)**.

Step 2: Candidate Item Pairs (\$C_2\$)

We combine items from \$L_1\$ into pairs and count support:

- **{Bread, Butter}**: 2 receipts (T1, T2) → Support = 66.7% (Passes!)
- **{Bread, Milk}**: 1 receipt (T2) → Support = 33.3% (✗ Pruned! Below 50%)
- **{Butter, Milk}**: 1 receipt (T2) → Support = 33.3% (✗ Pruned! Below 50%)

Only **{Bread, Butter}** becomes a **Frequent 2-Itemset (\$L_2\$)**.

Step 3: Rule Generation & Metric Audits

We extract rules from our frequent pair {Bread, Butter}:

- **Rule A: \$[Bread] \rightarrow [Butter]\$**
 - Support (Popularity): Support of {Bread, Butter} = $\mathbf{66.7\%}$
 - Confidence (Reliability): 100% (Passes $\geq 60\%$!)
 - Lift (Strength): 1.5x (Positive correlation!)
- **Rule B: \$[Butter] \rightarrow [Bread]\$**
 - Support: 66.7%
 - Confidence: 100% (Passes!)
 - Lift: 1.5x

Discovered Rule	Rule Popularity (Support)	Rule Reliability (Confidence)	Strength Multiplier (Lift)	Retail Recommendation
[Bread] → [Butter]	66.7%	100%	1.5x	Place Bread and Butter on the same aisle to drive convenience buys.
[Butter] → [Bread]	66.7%	100%	1.5x	Bundle them together in a "Breakfast Special" package promotion.

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